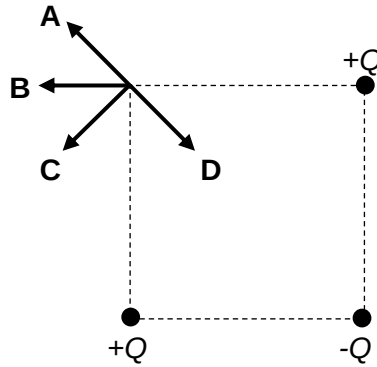


20 Three point charges, each of magnitude Q , are placed at the three corners of a square as shown in the diagram.

What is the direction of the resultant electric field at the fourth corner?



Answer: A

Due to symmetry, the net electric field at the fourth corner must be along the diagonal of the square in direction A or D.

Let E_1 and E_2 be the electric field strength due to each of the charge of $+Q$.

Resultant of E_1 and E_2 is

$$E_{\text{net of } +Q} = \frac{Q}{4\pi\epsilon_0 r^2} \sqrt{2}$$

Let E_3 be the electric field strength due to charge of $-Q$.

$$E_3 = \frac{Q}{4\pi\epsilon_0 (r\sqrt{2})^2} = \frac{Q}{8\pi\epsilon_0 r^2}$$

Therefore the net electric field strength due to all 3 charges is

$$E_{\text{net of } +Q} - E_3 = \frac{Q\sqrt{2}}{4\pi\epsilon_0 r^2} - \frac{Q}{8\pi\epsilon_0 r^2} = \frac{Q}{8\pi\epsilon_0 r^2} (2\sqrt{2} - 1) \text{ which is in the direction A}$$

